MERN STACK COURSE

Assignment: Personal Portfolio Website

Build a professional full-stack portfolio using the MERN Stack

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Deadline: [October 20, 2026] **Total Marks:** 10 **Type:** Individual **Submission:** GitHub Public Repository Link

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Objective

As future MERN stack developers, your online presence is your first impression to employers. In this assignment, you will design and develop a complete personal portfolio website from scratch using React.js (Frontend), Node.js + Express.js (Backend), and MongoDB (Database). Your portfolio must be deployed and fully functional — a living CV that showcases your skills, projects, and professional identity.

Required Sections & Features

1. Hero / Landing Section

- Animated intro with your name, role, and designation
- Typewriter effect or scroll-triggered entrance animation
- Call-to-action (CTA) button to Download CV
- Professional background or gradient design

2. About Me

- Short professional bio and your profile photo
- Location, current status (student/fresher/working)
- Animate section entry on scroll (e.g., fade-in, slide-up)

3. Skills Section

- Animated skill bars or radial/circular progress charts
- Group skills by category: Frontend, Backend, Tools & Technologies
- Hover effects and tooltips showing proficiency level

4. Projects Showcase

- Minimum 3 projects displayed as interactive cards
- Each card: project name, description, tech stack badges
- GitHub repository link and live demo link for each project
- Filter/sort by technology (optional but adds marks)

5. CV / Resume Download

- Allow users to download your resume in PDF format
- Allow users to download your resume in Word (.docx) format
- Files must be served via a Node.js/Express.js backend route

6. Education & Experience Timeline

- Vertical animated timeline layout

- Show degrees, institutions, years of study
- Include internships or work experience if applicable

7. Certifications & Achievements

- Display certificates from Coursera, NPTEL, LinkedIn Learning, or any platform
- Show certificate image or link to verified certificate
- Include any hackathon participations or awards

8. Contact Form (Backend Connected)

- Functional HTML contact form with Name, Email, Message fields
- Form connected to Express.js backend API route
- Messages stored in MongoDB using Mongoose
- Optional: send email notification via Nodemailer upon form submission

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Technical Requirements Checklist

All of the following must be implemented to receive full marks:

- React.js frontend with React Router (single-page application)
- Node.js + Express.js REST API backend
- MongoDB + Mongoose for storing contact form submissions
- Some animations (Framer Motion, GSAP, AOS, or CSS animations)
- Fully responsive design — must work on mobile, tablet, and desktop
- Dark mode / Light mode toggle
- CV available for download in both PDF and Word (.docx) format from backend
- Deployed live — Frontend on Vercel or Netlify, Backend on Render or Railway
- Environment variables used for sensitive data (.env file NOT pushed to GitHub)

Marking Rubric

Criteria	Details
UI Design & Responsiveness	Clean layout, mobile-first design, consistent color theme, dark/light mode toggle
Animations & Interactivity	Smooth scroll animations, animated skill bars, typewriter effect, hover states
MERN Stack Integration	React frontend + Express API + MongoDB contact form working end-to-end
CV Download (PDF + DOCX)	Both formats downloadable via a working backend route, correct file served
Projects & Content Quality	Minimum 3 projects with GitHub + live links, real meaningful descriptions
Deployment	Frontend and backend both deployed live and accessible via public URL
Code Quality & GitHub	Organized folder structure, meaningful commits, README.md, .env not exposed

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Submission Instructions

Submit your GitHub public repository link in Google Classroom by the deadline.

Your repository must include:

- Complete source code with /client (React) and /server (Node/Express) folders
- A README.md file with: project description, live deployment link, and local setup steps
- All dependencies listed in package.json files
- A .gitignore file that excludes node_modules and .env

> ⚠️ **IMPORTANT:** Never push your .env file to GitHub. Always use .gitignore properly. Repositories with exposed MongoDB URIs, API keys, or passwords will have marks deducted. Keep your credentials secure.

Good luck! This portfolio is not just an assignment — it is the first page of your professional career.